

STICK IT TO THE MAN IN THE MIDDLE

Man in the Middle (MitM) attacks have been around for a long time. The method involves intercepting communications between machines, and fooling both parties into believing that the data is from a legitimate source. There is a potential for eavesdropping and picking up sensitive information, as well as modifying the data being relayed for criminal purposes.

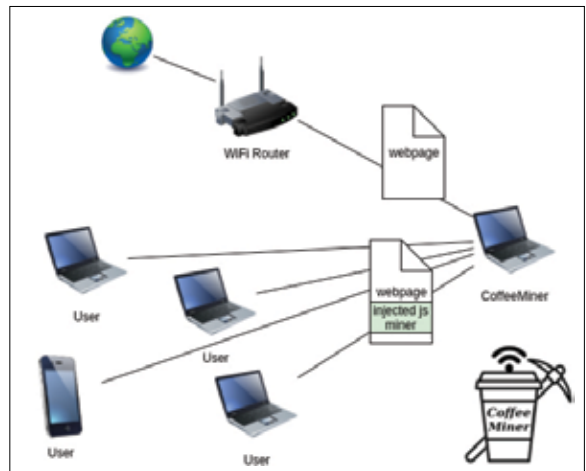
One of the biggest security concerns is public WiFi hotspots. The approach works by capturing a number of packets going back and forth between the target device and the access point, capturing the details of the initial handshake between the target device and the access point, or using ARP spoofing to fool the access point and the target machine. The method can be used to capture credentials. Another approach a malicious attacker can use is to set up their own access points with nomenclature similar to those commonly used in public, or known to exist at a particular location. It is best to avoid using public hotspots for critical and sensitive tasks. Flawed implementation of MitM protection measures in apps, including TunnelBear, a popular VPN service, and HSBC's banking app, actually make them more vulnerable to MitM attacks.

MitM attacks are not just for attempting to steal credentials or sensitive documents from the user. An attacker can literally make money by just sitting on an open access point. The idea is to use a cryptocurrency miner that can use the computing power of all the devices connected to the access point. The MitM attack on all the connected devices is used to inject custom JavaScript code in the HTML pages requested by the victims. Within the script is a cryptocurrency miner. The attacker can execute a

simple command on their devices, which will form a LAN network among the compromised devices, all of which will start mining cryptocurrency. Such an attack can be executed through publicly available open source tools, such as Coffeeminer. Research from AdGuard has indicated that the devices of over 500 million users are made to mine cryptocurrency through browser windows every month, without their knowledge.

The MitM approach is actually used by some companies to secure

their cyber assets. Organisations inspect traffic between machines, to make sure that the data is being legitimately transferred. The approach has been proven to work for protecting organisational networks, as well as cloud storage. According to a prediction by Forcepoint Security, malware will get



How Coffeeminer works in tandem with a MitM attack.

increasingly aware of MitM protection in 2018. Once the malware realises that the traffic is under analysis, it will stop transmitting the data. ■

PRIVACY? WHAT PRIVACY?

By now, you must have seen enough tech dystopian movies where the state has become a surveillance state employing all kinds of technology to keep an eye on its citizens. While the terminator might not be headed for your home, a lot of technology is already in there, and out here, that is perfectly capable of spying on you.

Remember the spying fiasco with the Google Home Mini? One where it was listening in on users and uploading the data to Google's servers, regardless of the activation phrase being used? Expect that to happen a lot more as everyone gets onto the smart speaker bandwagon, and more alternative products are released hastily in existing ranges. Even if that doesn't happen out of the box, it won't exactly be difficult for a hacker with malicious intent to make a smart speaker listen in on you. In fact, that could be done even before 2017. They could even make it play random songs or tracks, or voice commands to control another nearby smart speaker (<http://dgit.in/boseroll>). It helps that platforms like Alexa and Google Assistant are completely open and lightweight.